

✓ Get technically strong in system-side computing in 30 hours

☀ WEEK-WISE PLAN:

(-Learn Touch Typing at all cost)

✓ Week 1: Understand Computer Hardware (6–7 hours)

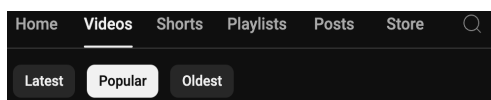
Objective: Build clear mental models of how the computer physically works.

📖 Topics:

- CPU, RAM, HDD/SSD, GPU – what they do
- Buses (data, address), clock speed
- Motherboard & chipsets
- I/O devices and ports
- (How a computer boots), (understanding and knowing to work on BIOS/UEFI to OS)-- **IMP**

📺 Suggested YouTube Videos:

-  **How does Computer Hardware Work?**  [3D Animated Teardown]
-  **Computer Components For Dummies**
-  **BIOS and UEFI As Fast As Possible**
-  **How to Set Up a Home Media Server**
- Keep Watching videos from these Playlist - [Techquickie - YouTube](#) [watch Popular videos in this channel or something u find interesting]



🔨 Practice:

- TRY TO DUAL BOOT ON UR
- Watch teardown videos of a PC/Laptop (just YouTube)
- Explore CPU-Z and Speccy on Windows or Iscpu, Isblk on Linux

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✓ Week 2: Get Confident with Operating Systems (8 hours)

Objective: Understand how an OS works and interact with it at command level

Topics:

- OS role: process, memory, file, device management (Surface level)
- Boot process, kernel, user space file systems (FAT, NTFS, EXT) (Surface level)
- Linux vs Windows basics , spend few hours working in linux (kali linux is suggested)

LEARN MAJORITY TERMINAL COMMANDS IN WIN AND LINUX:

WIN:-

-  **40 Windows Commands you NEED to know (in 10 Minutes)**
-  **30 Windows Commands you CAN'T live without**
-  **how to HACK a password // Windows Edition**

LINUX:-

-  **60 Linux Commands you NEED to know (in 10 minutes)**
-  **37 INSANE Linux Commands you NEED to Know in 2025**
-  **The Most Dangerous Linux Commands**

Suggested Resources:

- [Linux Journey](#) – to practice Linux basics in browser
- [NetworkChuck Youtube Channel {The Best}](#)
- [DistroTube's Linux CLI tutorials](#) – highly practical {dont stress to much on understanding everything at once }

Practice:

- Use a VirtualBox/WSL or real Linux OS (Ubuntu recommended)
- Use terminal: ls, cd, pwd, top, ps, kill, nano, chmod, sudo
- Try shell scripting basics (.sh files, bash)



Week 3: Networking Basics (7 hours)

Objective: Get a clear picture of how devices communicate



Topics:

- IP address, MAC address (DNS, DHCP, NAT)
- HTTP/HTTPS, FTP, SSH, TCP vs UDP
- OSI Model – just 7 layers and what they mean
- Ports & firewalls
- Network devices: switch, router, hub, modem



Suggested Videos:

- [How does the INTERNET work? | ICT #2](#)
- [How Data Travels Through Wires: A Simple Guide](#)
- [what is an IP Address? // You SUCK at Subnetting // EP 1](#)
- <https://youtube.com/playlist?list=PLsep1uQfemNFvT57-DOjzSnGspJv8efpS&si=IBITGyYlq-G4BtcN>



Practice:

- Use ping, traceroute, ipconfig/ifconfig, netstat, curl
- Connect 2 virtual machines using host-only adapter
- Try basic SSH: ssh user@ip



Week 4: Integration + Practical Usage + Linux Projects (8 hours)

Objective: Put everything into practice by doing 1-2 mini projects or labs



Projects / Labs:

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- Install Ubuntu in VirtualBox and explore:
Installing packages with apt
Writing a simple shell script
Creating and killing processes
Monitoring system resources
- Create a home network map
 - IP addresses of devices
 - Use nmap to scan your own network
- Set up SSH from one system to another
- Use Wireshark to see packet flow (optional but fun)

Suggested:

-  you need to learn Virtual Machines RIGHT NOW!! (Kali Linux VM, Ubuntu, Windows)

Tools/Apps Required:

- VirtualBox + Ubuntu ISO
- LinuxJourney.com or Termux (Android)
- Optional: Wireshark, nmap, SSH, Notepad++, VS Code

Review Plan

- Ask her to **maintain a log or summary doc**
- Let her do **one peer explanation every week** (teaching builds mastery)
- After 4 weeks, she should be able to:
 - Open a terminal and feel confident
 - Explain how her computer starts up
 - Do basic network diagnostics
 - Understand what processes and threads are doing

-ARADHYA